close to the busbars, between the busbars and each motor oil switch, each set being contained in an imitation stone cell.

The greater part of outgoing feeders are supported by iron brackets on the walls of the houses, just below the roof, and as all the lamps in the streets are suspended on transversal suspensions, the wire runs down the wall of the house, along the suspension to the lamps and returns by the same way.

In the squares and parks the lamps are on poles of the usual type, and the feeders are underground.

The substation lighting is done by means of metal filament lamps on an alternating current circuit, the low tension being obtained from a small transformer on one phase. In case the current on the line should fail, current from a battery of accumulators is mechanically switched onto the lamps.

The battery, which is small, being only for 40 amp-hours, is in a room in the basement, and only used in special cases for lighting the station, and is chiefly used for the laboratory.

The maintenance of the whole system is fairly cheap; the substation is only in service during the night, that is, from about 6:30 p. m. to 6 a. m. in the winter and 8 p. m. to 3 a. m. in the summer, after midnight there are only half the number of lamps burning.

The substation staff consists of one engineer, two switchboard attendants, four helpers, one storekeeper, two skilled workmen for lamp repairs, four workmen for general repairs and 11 laborers. Besides these there are 22 arc lamp attendants. Each lamp is visited once daily.

All the machinery and switchboard material was supplied by the General Electric Company, of Schenectady, and the cables by the Pirelli Company, of Milan. The erection of the substation was all carried out by the municipality's staff.

## ELECTRICAL TRANSMISSION OF MUSIC.

### Developments in the Cahill Telharmonic System.

OUR readers are familiar with the general features of the Cahill Telharmonic system for generating currents of the proper frequency and amplitude at a central station, and superposing and transmitting them to numerous distant points, for the purpose of producing music by means of electromagnetically vibrated diaphragms. During the past three years important changes have been introduced in the equipment, and the results now being obtained with a new and larger plant, which is being completed in the Cahill laboratory at Holyoke, Mass., are even better—in some ways strikingly better—than those secured with the equipment formerly in service in New York City. The power and breadth of effects and the range of expression have been greatly increased. The mechanism has been simplified and the keyboard can now be manipulated by any skilled pipe-organ or piano player after only a few trials, with results much more pleasing than they can obtain with the instruments upon which they have spent years in study and practice.

But, notwithstanding this, the instrument has a new technique, a thorough mastery of which is not easily obtained. A well-known musical critic puts it thus: "Any good pianist, with a few hours' practice, can produce some very pleasing effects on the new Telharmonium. But it would take a great musician 10 years to master thoroughly its technique."

A description of the electrical features of the New York equipment was contained in our issue for March 10, 1906, and Jan. 5, 1907. In the present article will be given an outline of the more interesting of the improvements that have been introduced in the new equipment.

Before taking up the improvements in detail it may be of interest to the reader to know what was desired in building the new plant. The ability of the New York plant to produce a wide range of orchestral timbres and to control the loudness of the note at every instant, by the human touch, are known to all who saw and heard it or to those who have read the previous descriptions in this journal. But it fell short of the inventor's ideal in ways that will be understood from the explanation below.

In the New York plant, each switchboard represented a voice, but to produce the full, rich, many-voiced effect of a fine orchestra, quite a number of switchboards are required. Dr. Cahill's design contemplated five switchboards, with a large number of generators, for the New York plant, but the great expense of that pioneer work compelled a large reduction in the number of generators and the omission of three of the five switchboards. The distinctive tone and timbre of the music were there, but the full orchestral effect which the inventor desired was not. In this important particular, then, the plant fell below the ideal, not from any defect in the invention, but from necessary requirements of economy; notwithstanding all of these economies the original New York plant cost almost $200,000.

Another of the difficulties with the original New York plant was that on account of the interactions or reactions of the electrical circuits, the more keys of any one keyboard were depressed at the same instant (that is, the fuller the chord) the less loud each individual note became; and a point was easily reached where the loudness and strength of a chord were less than that of a single note. This defect is known at the laboratory and among the musicians as robbing—that is, one note robbed another of its strength. It was never serious so long as only one note like a single orchestral "voice" was played on each keyboard; it was scarcely noticeable with two notes on a keyboard. When the number of keyboards is small, or when one musician plays alone, it is extremely desirable to be able to take a full chord with either hand, as on the pianoforte or organ. In the new Telharmonium the circuits, switches and transformers have been reorganized in such a manner that full chords can be played on the same manual and switchboard without any appreciable diminution in the strength of the individual notes, through the reaction of the circuits. When one considers that at times 50 or 60 circuits, corresponding to different harmonics of the different notes in the different switchboards, are acting upon the line and reacting on each other simultaneously, the manner in which each tone and overtone maintains its independence is quite remarkable.

Before Dr. Cahill had delivered the plant to the New York Company, to whom it was leased, he determined, if possible, so to improve the whole apparatus as to reduce the very great cost which the switchboard work involves. In the New York plant, the loudness of the note was controlled by the touch of the hand, and effects very similar to those of the bow were produced, but with short notes and rapid movement it was difficult to co-ordinate the dynamic control with the control of the pitch without much skill and experience on the part of the player. In the new plant these difficulties have been greatly reduced and some have been entirely eliminated.

Persons who have heard the Telharmonium, or dynamophone, as Dr. Cahill prefers to call it, have usually been surprised at the round and powerful tones which, under the influence of its electrical currents, were produced from an ordinary telephone diaphragm. But here also there were limitations which imposed burdens upon the performer. An ordinary telephone diaphragm responds well to powerful electrical vibrations so long as only one note is used. It also responds well for many of the chord effects, but there are other chord effects, particularly close chords in the bass, which if taken *fortissimo*, produce an unmusical sound—almost a growl. Dr. Cahill found two remedies for this defect. One was to produce one bass note from one diaphragm and line and another bass note from another diaphragm and line. This method obviously required a multiplication of lines and receivers, thereby involving a great increase in expense. Another remedy was to open up the chords in the bass. This method, however, involved the rearranging of the music in many cases. These difficulties have been completely overcome by a new receiver described below, which was designed by Mr. Arthur T. Cahill, who is associated with his brother in the electrical work.

All who have worked with high-frequency alternators have learned the great difficulty of producing any considerable output at frequencies of more than, say, 1000 cycles per second. This was another point in which the New York plant fell below the inventor's standard and in which the superiority of the new plant is very striking.

Good intonation was obtained at the plant in New York, but this perfection of intonation was produced at the cost of much mechanical complexity, an expensive generating set, expensive switchboards and a new and difficult keyboard, which the musician had to study for a considerable time before using. The resulting musical gain was worth what it cost, but efforts were directed to reduce the cost and eliminate, as far as possible, all the difficulties involved. This move seems to have been in the right direction, for the difficulties to the musicians, which resulted from the complicated keyboards of the New York plant, often led them to neglect the best powers of the instrument in the effort to accomplish something musically with little experience in playing and little time to practice.

So soon as the so-called New York plant had been leased to the New York Company, and even before it was installed in New York, Dr. Cahill began another elaborate series of experiments with musical alternators and switchboards. A switchboard of greatly increased musical powers was built in 1906 and 1907, but, like the preceding switchboards, at a great expense. It was appreciated that one important way to reduce the expense of the switchboard work (paradoxical as this may seem) was by improving the generating set. Many alternators, of more or less novel construction, were built and tested in 1906, 1907 and 1908. Experiments were made also on high-speed journals and gearing, and much new work belonging to the switchboards proper was carried on at the same time. In 1908, the necessary preliminaries having been settled, the work on the new plant was pushed and the result is now apparent.

The new plant is much larger than the original New York plant. That plant had eight sections or panels of generators. The new plant has 24 sections in operation, with provision on the main frame and in the gearing system for others to be added later. The new sections are smaller and contain a smaller number of generators, but the individual generators, or at least those of high frequency, are much more powerful. In fact, the generators of the very highest frequency have many times the output ratings of the corresponding alternators in the New York plant. Almost the only feature of the New York equipment retained, without change, in the new plant, is the type of alternators, which are inductor machines. Even here, however, there have been changes in the details of the winding and in the shaping of the pole pieces, etc., the result of a long series of experiments.

The generating set is much larger and occupies a much larger space than the generators of the New York machine; on the other hand, the switchboards have been greatly simplified and cheapened and occupy much less space.

Each panel of generators is mounted upon a steel shaft about 8.5 in. in diameter, which runs in a cast-iron shell. These shells are assembled upon a steel main frame. The main frame is approximately 65 ft. long, 13 ft. wide and is built of 20-in. steel beams, mounted upon masonry. Great strength is apparent in the framework and foundations. The result is that the 24 high-speed shafts, before mentioned, each carrying a different group of inductors, run quietly and without any perceptible vibration.

As one might naturally expect, when 24 shafts are to be connected instead of eight, the gearing is much more complicated and expensive than in the New York plant. All the gears are completely cased and run in oil or lubricating grease. No explanation of the details is possible except with the aid of elaborate drawings, which at this stage would be of little interest to the public.

The generating part of the plant is so crowded in with the switchboards and their thousands of wires, on the one side, and the heavy machine tools of the laboratory on the other, that it was impossible to obtain a good photograph showing the whole.

Fig. 1 illustrates a small part of the generating set, showing some four or five of the 24 panels of generators. Fig. 2 shows one of the keyboards (the only one that is yet complete) with the musicians at work. Fig. 3 illustrates a portion of a switchboard which is being connected into service.

The steel shafts, carrying the inductors, are mounted on ball-bearings of chrome-nickel steel, made in Germany. This feature alone would be interesting to the intelligent central-station man. Here is an immense machine, with perhaps the most complicated gearing ever built, and 140 alternators, all running at high speeds, on ball bearings. The saving of power in the bearings alone is estimated at more than 150 hp.

The switchboards, even in their improved form, are of so complicated a nature that little can be told about them in a few words and little shown by a photograph. Detailed schematic drawings would be necessary, by which alone the mutual relations of the great number of circuits involved can be

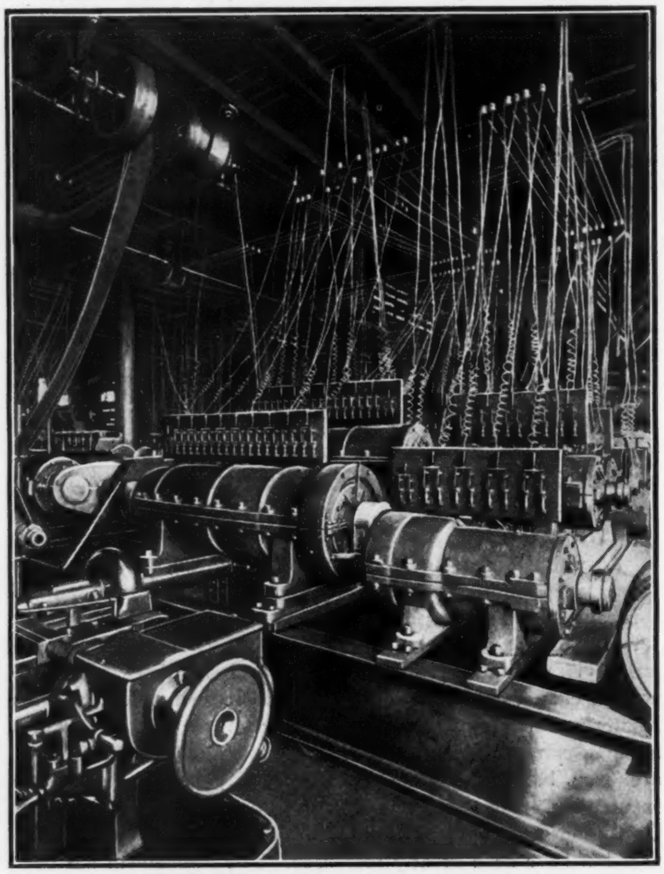

Fig. 1—Generators of Telharmonium.

understood. Central-station men who have struggled with the difficulty of making two alternators run in parallel, will appreciate what it means to work 140 alternators, in parallel, without hitch or trouble in any of the switchboards.

The new switchboards are characterized by much greater simplicity than those of the New York machine. Equally good effects are obtained in some cases with one-sixth the number of switches and circuits. In others, the electrical system has been reorganized in such a way that a great reduction in the number of transformers is effected. Moreover, the new switchboards are organized in such a manner as to make the musician's work easier—to improve and increase the powers of expression on the instrument. In particular, the features of touch-control, described in Dr. Cahill's first patent, have been, as before said, much improved, and in such a manner that the expense of the plant and the difficulties of the technique are reduced at the same time that the powers of expression are increased. Some new expression devices, also, have been added at the keyboards.

Mention was made above of the great increase in the power of the high-frequency alternators—the alternators for the high-pitch notes. This increase is due partly to changes in scale, partly to higher speeds, partly to smaller air-gaps and improved windings.

In Dr. Cahill's Washington plant, built in the late '90's, the transformers used resembled the college induction coil or Ruhmkorf coil, more than the transformers used in electric lighting work. In the New York plant, the transformers were more nearly like the usual static transformers of electric lighting. In the new work the transformers are of the original

Fig. 2—Keyboard of Telharmonium.

type. The musical conditions are so far different from those of lighting circuits that the open-circuit transformers seem to be preferable to the closed-circuit transformers.

The new receiver, by which the electrical vibrations are converted into sound waves, is very different from that of the ordinary telephone. It has a much larger and heavier diaphragm—about 10 times as large and three times as thick. A more powerful magnet system is used and the disposition of the field and coils is strikingly different from that in common use. The correct formation of the air passages, as well as the length and depth of the horn itself, have received much attention from Mr. Arthur T. Cahill. Partly as a result of the increased size and weight of the diaphragm, partly from the

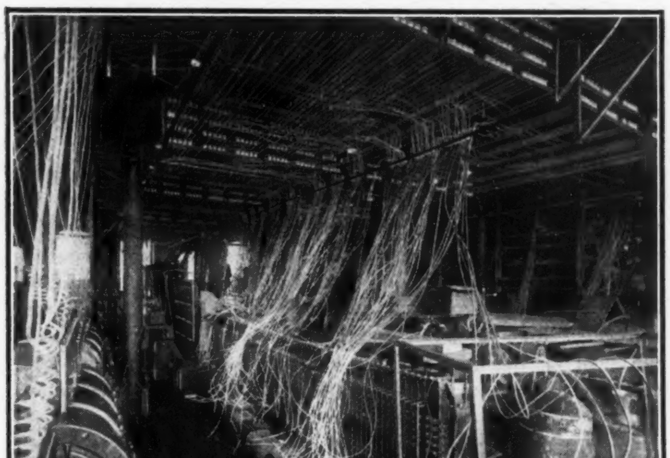

Fig. 3—Switchboard in Process of Erection.

greatly increased strength of the magnetic system, partly from the novel disposition of the same and the novel shape of the air passages, a most striking effect is produced. A single diaphragm responds powerfully to any sound, from the deepest bass to the highest treble of the machine; to a pure tone or a highly composite tone; to a single note or to a full chord. The volume and purity of the sound is very striking.

From the generating equipment described above may be obtained currents for superposition at any available number of switchboards by any desired number of performers at the manuals. In general, there would be a separate manual for each switchboard. The switching arrangements are such that at each manual the ground tone and over tones for each note can be superposed in any chosen proportion to produce any desired quality of tone; for example, to imitate the 'cello, the oboe or the clarinet at will. At each manual the performer makes the adjustments for the quality desired, just as is done with a pipe organ, but the range of adjustment is far greater with the Telharmonium. With a pipe organ, each stop sounds with its full strength or not at all; at a wind pressure less than that on which it is "voiced," it would sound out of tune. With the Telharmonium, on the contrary, any stop can be drawn more or less, so as to sound softly or loudly, accordingly as it is opened more or less, and it always sounds perfectly in tune, whether soft or loud.

The inventor, Dr. Thaddeus Cahill, has expressed the conviction that the most important musical effects of a large and fine orchestra can be produced electrically by the use of, say, 10 manuals, with one performer for each and a sufficient set of generators to furnish all of the necessary harmonics. The music, of course, of this electrical orchestra would not be exactly the same as that of the orchestra of bows, woodwinds and brass. It would certainly be better in some ways; it might not be so good in others. It is hard to find two pianos or two violins that are exactly alike. And the differences between the electrical violin tones and the tones of a Stradivarius will probably be greater than the difference of tone between a Stradivarius and a Guarnarius. The electrical music, while capable in a way of imitating all sorts of instruments, has yet its own strongly-marked individuality. It is capable of producing tones of all sorts of timbres, by combining the different harmonics or overtones and the ground tone in different ratios. It can produce different orchestral "voices" at the same time, by employing one switchboard and keyboard for one voice and another switchboard and keyboard for a different voice. It possesses remarkable powers of expression which it gives to the performer, equaling those of the violin in respect of soft-and-loud effects, with that control over the loudness of the note at every instant which only the bow possesses, while excelling the violin in chord capacity or double-stopping power and range of timbre-control. It has a remarkable goodness of intonation, which is equal, if not superior, to that produced by the finest quartet, and much superior to that produced by a full orchestra, with brass, wood-winds and strings. All of the above features make it clear that the goodness of the music of a telharmonic plant will depend only upon the number of generators and switchboards—the capital invested—and the skill of the players.

A third switchboard is being connected in, as Fig. 3 shows, and others will be added later. With only two switchboards in service it is yet possible to obtain a range of quality much superior to the most elaborate pipe organ or at least to obtain much more effective tones than the best organ furnishes, with a range of expression equal to that found in orchestral music.

In comparing the new Telharmonium with the piano, violin or organ, one should think of the millions of those instruments that have been built, of the thousands of years of evolution that lie behind them, of the vast armies of teachers and players in all countries of the world, of whom a few of the best only, like Paderewski, Hoffman, Ysaye, Kreisler and Elman, stand forth, with their great genius and their long training, to show what can be done with those instruments. One should remember that the latest is only the third Telharmonium that has been built and that the players who used it had only a few weeks' experience with its new technique—the best of them, the pianist, Mr. Paul Fishbaugh, of Washington, D. C., only four days. When these facts are borne in mind, an electrician, at least, can be excused for thinking that the day is near when the superiority of the electrical music to the music of string and pipe will be as obvious and incontestible as the superiority of a fine automobile to a farmer's wagon. But the electrician will not be alone in this view. All the musicians who have seen the new Telharmonium are in love with it. And almost every person who hears it considers it as the music of a future that ought to begin at once.

The new plant has cost Dr. Cahill about $200,000 to date.

Apart from the striking improvement in the musical effects which have been made, the chief merit of the new plant lies in the great reduction effected in the cost of the switchboard work. The new generating set, with its gearing, is, as before said, even more expensive than the generating set of the New York machine, but it is also a much better, more powerful and more nearly complete generating set. The great importance of the reduction made in the cost of switchboards results from the fact that numerous switchboards are necessary in order to produce a full orchestral effect. With the improvements made these switchboards can be added at a cost that is not prohibitive.

The inventor claims that for $300,000 a powerful plant, with eight good switchboards, can be built at the present prices of labor and material, while three good switchboards and enough generators to make a respectable beginning in a city having 200,000 or 300,000 inhabitants can be furnished for $125,000. With such reductions in cost, combined with the great improvements in the musical effects before described, the electrical music ought to come into general use at an early date.

In all of the electrical music work Dr. Cahill has, from the beginning, had the valuable co-operation and assistance of Mr. Arthur T. Cahill, and the new receiver, which adds so much to the musical effect, is the latter's work. A third brother, Mr. George F. Cahill, has given his co-operation also.

The improvements made in the new plant may be briefly stated to consist in a great improvement in musical effects, combined with a great reduction in the cost of the plant. This reduction in cost arises chiefly from the greatly increased power of the high-frequency generators and various improvements in the generators and their gearing, a great simplification in the switchboard work, and the improved receivers by which a single receiver and a single pair of lines is made to do the work which formerly required several receivers and several pairs of lines. The improvements in musical effects arise chiefly from the greater strength of the highest notes and of the higher harmonics, the improvements in the expression devices, and the simplifying of the keyboard and switchboard work, making it much easier for a performer, familiar with the pianoforte or pipe organ, to play upon the new Telharmonium.

## STREET ILLUMINATION.

### Photometric Tests of Street Lighting in Budapest.

By Francis Jehl.

As the street lighting in Budapest, which was designed and erected by Mr. Etienne de Fodor, general manager of the Budapest General Electric Company, had given entire satisfaction in every respect for nearly a year, Mr. de Fodor decided to have a series of photometrical tests made, which would tend to augment the good results so far obtained. This decision was communicated to the municipal authorities, who were pleased with the idea, and promised to be represented at the tests by several members of their technical staff. In order to impart an impartial character to the tests, it was decided to call in an outside expert to act in conjuncion with a local committee. As expert, Mr. William Wissmann, of Berlin, was chosen; he is also a member of the German Standardization Commission, and who has had a large experience with arc lamps, street lighting and everything pertaining thereto. The names of the members of the local lighting committee are as follows: E. de Fodor, general manager Budapest General Electric Company. D. Jasz, technical councillor to the State; A. Söpkez, professor at the Budapest Polytechnic; A. Heuffel, chief of the Building Department; J. Rozsenyi, chief city chemist; E. Modrovich, assistant city chemist; B. Semsey, chief engineer of the city; J. Vukasinovits, chief engineer of the city; A. Stromszky, general manager of the Siemens-Schuckert Works; F. Jehl, engineer of the Budapest General Electric Company; C. Járolimek, engineer of the Siemens-Schuckert Company; J. Nitsche, R. Fuzy, G. Jaczo, L. Ban, as assistants. The photometer used was a Weber type; made by Schmidt & Haensch, of Berlin, while the standard was a small tungsten lamp that was carefully standardized in Berlin, and received its current during the tests from a small portable storage battery. The standard lamp was adjusted by varying the resistance of the circuit in accordance with the indications of an ammeter. It may be here mentioned that it has been found far more practicable and exact to regulate a standard lamp with an ammeter than with a voltmeter connected across its terminals, since the error caused by a faulty contact in a voltmeter circuit can produce misleading results.

The diffusion plate from which the light intensity was measured, consisted of a carefully ground surface of plaster of paris, which was treated with an oxide of magnesia.

It has been found that such a surface gives the least possible error, while white paper, ground glass or simple plaster of paris and other kinds of surfaces often used are liable to errors that may be as great as 50 per cent and more. In connection with the above subject, it may be mentioned that researches have been carried on in the laboratory of Siemens-Schuckert for over a year, and that the outcome of the work

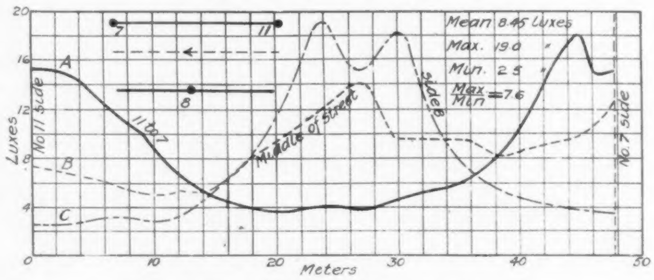


Fig. 1—Distribution Curves, Arc Lamps.

was that the plaster of paris-magnesia plate has been found to excel all other kinds. The height of the plate from the level of the ground was fixed at 1 m and from this height the intensities were measured. It is also said that the German Standardization Commission will propose the 1-m height as the standard to be taken in photometrical measurements. In laying out the lines upon which the measurements were to be carried out, the local committee decided to measure the various intensities between two arc lamps in a direct line on one side of the street, then a corresponding one on a parallel line in the middle of the street, and similarly on the other side of the street, where one arc lamp was situated. The lamps in the Rakoczi street are placed zigzag, first on one side of the street and then on the other. Sufficient data were obtained to give a very good idea of the distribution of the light. All precautions were taken to guard against vagabond rays; the tests were commenced at midnight and lasted generally until 5 o'clock in the morning, shops being thus closed, and the street gas lamps were turned off during the time of testing.

The results of the first night's work are shown in Fig. 1; above the curves is shown a sketch of the area tested and the position of the arc lamps. Curve A gives the various intensities from arc lamp No. 11 to arc lamp No. 7 in a direct line, while curve B gives the intensity in luxes for the middle of the street from side 11 toward 7, as shown in the dotted line above the curves. Curve C gives the light intensity along the side where arc lamp No. 8 is situated. The width of the street was about 29 m, while at arc lamp No. 11 a small square is situated, and thus from this side there was no reflection as buildings sometimes offer. From the curves it will be seen that the mean intensity is about 8.45 luxes, the maximum 19,